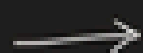




Monday, November 17, 2025

9:10 AM

Introduce the problem statement



Agenda



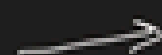
(what we will be covering in the presentation)

Background info

(introduce the ruleset, the sensor modalities, and the current state of Arkenis)

→ Regulations we have to consider

(our chosen 5 locations, what regulations we fail with our current sensor suite)



Our MVP

(The sensors on the market, our placements, the FOV, total cost, power, mileage, etc.)

→ ODD $\frac{1}{3}$ self certification
(a lot of info will be hidden slides)

Logos



aUToronto

R2Y5 0-0-0 Challenge - Sensor Suite Optimization



**By: Prithvi Seran, Amanda Liu,
Forest Li, Nevan Kho, Ailing Ji**



For the R2Y5 competition, all teams had to engineer a sensor suite for the car.



There's a problem...



**These sensor suites end up
being impractical!**



Optimize the sensor suite of your car.



R2Y5 RULESET

Extra rules we have to consider in addition to our problem statement

Compliance with specific regulations

FMVSS111, FMVSS127, FMVSS208

Compliance with regulations in 5 states/provinces

California, Michigan, Texas, Arizona, Ontario

Benchmark analysis with another robotaxi

We have chosen Waymo and Tesla



Assumption: We will only
be optimizing for
regulatory requirements
and road legality.



AGENDA



Background Information



Sensor Modalities



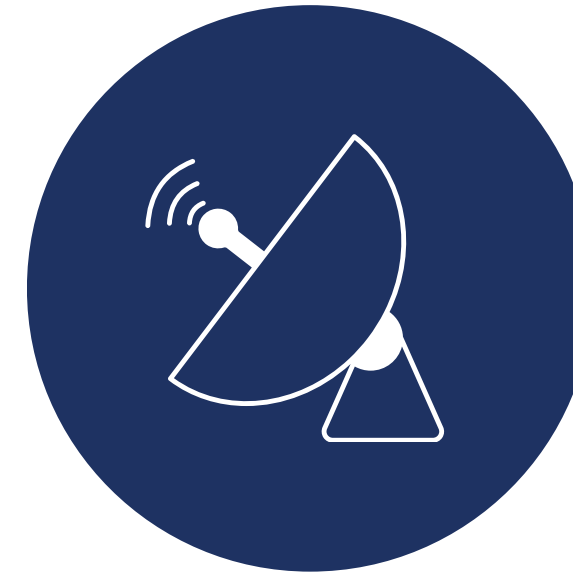
Cameras

2D image capturing for recognizing road signs, lane markings, and colors



LiDAR

Detects obstacles, road features, and pedestrians using 3D point clouds

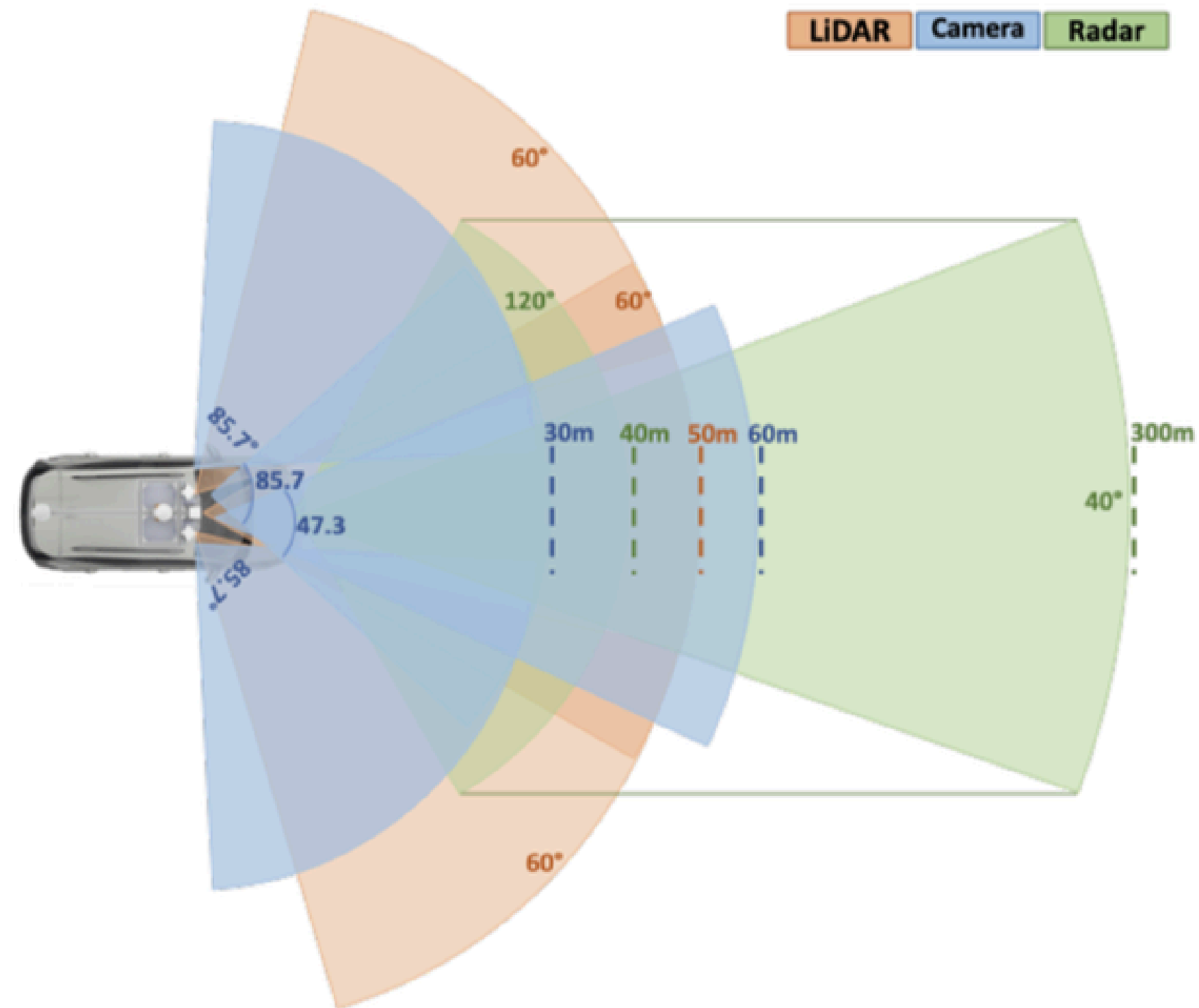


Radar

Tracks objects and their speed using doppler effects



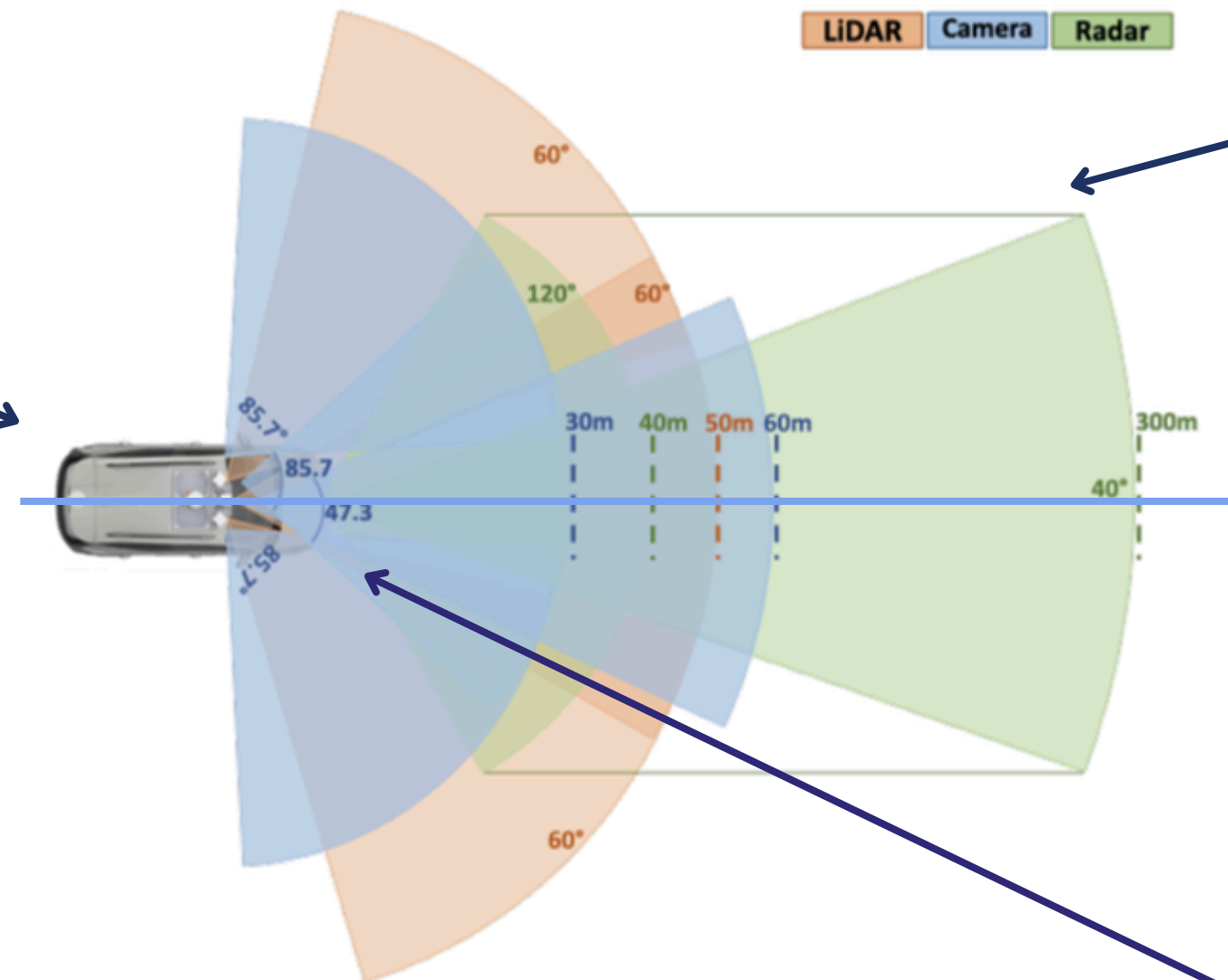
Our R2Y5 Sensor Suite



A lot of room for improvement...

Completely no coverage on the back of the car

300 meter range is not needed



Lots of overlap



THE PROBLEM

An Impractical Sensor Suite



01

Sensor Coverage

Lack of cov



02

The Problem 02

It is mostly presented
before an audience.



03

The Problem 03

It is mostly presented
before an audience.



THE SOLUTION

Presentations are tools that can be used as lectures, speeches, reports, and more. It is mostly presented before an audience.



The Solution 01

It is mostly presented before an audience.



The Solution 02

It is mostly presented before an audience.



The Solution 03

It is mostly presented before an audience.



Regulatory Compliance

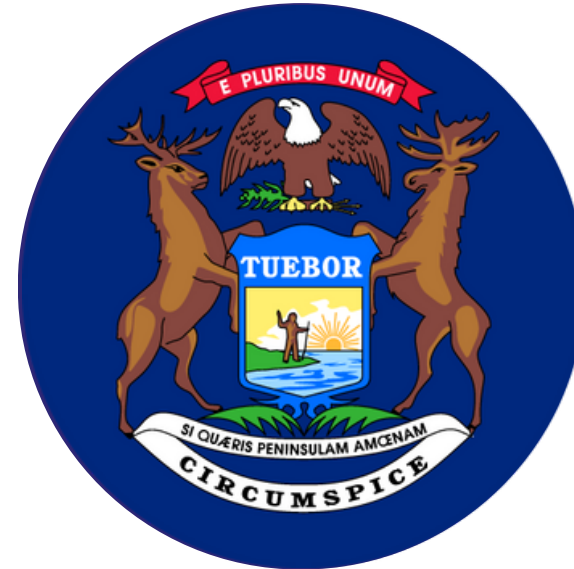


OUR STATES & PROVINCE

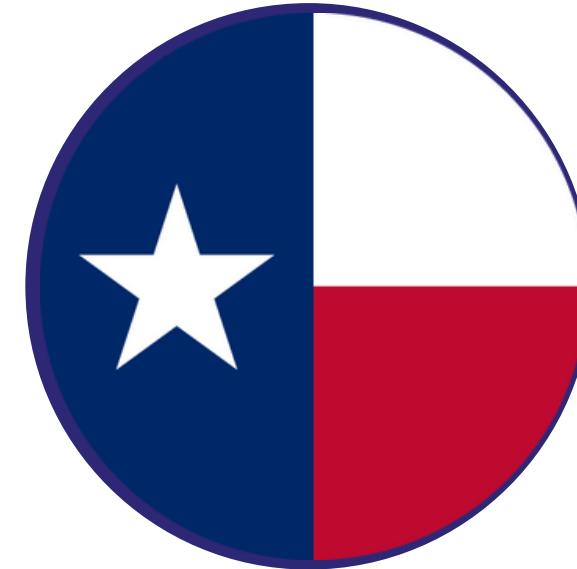
Required
States



California, USA



Michigan, USA



Texas, USA

Additional 2



Arizona, USA



Toronto, CAN



aUToronto

OUR SCENARIO & ASSUMPTION

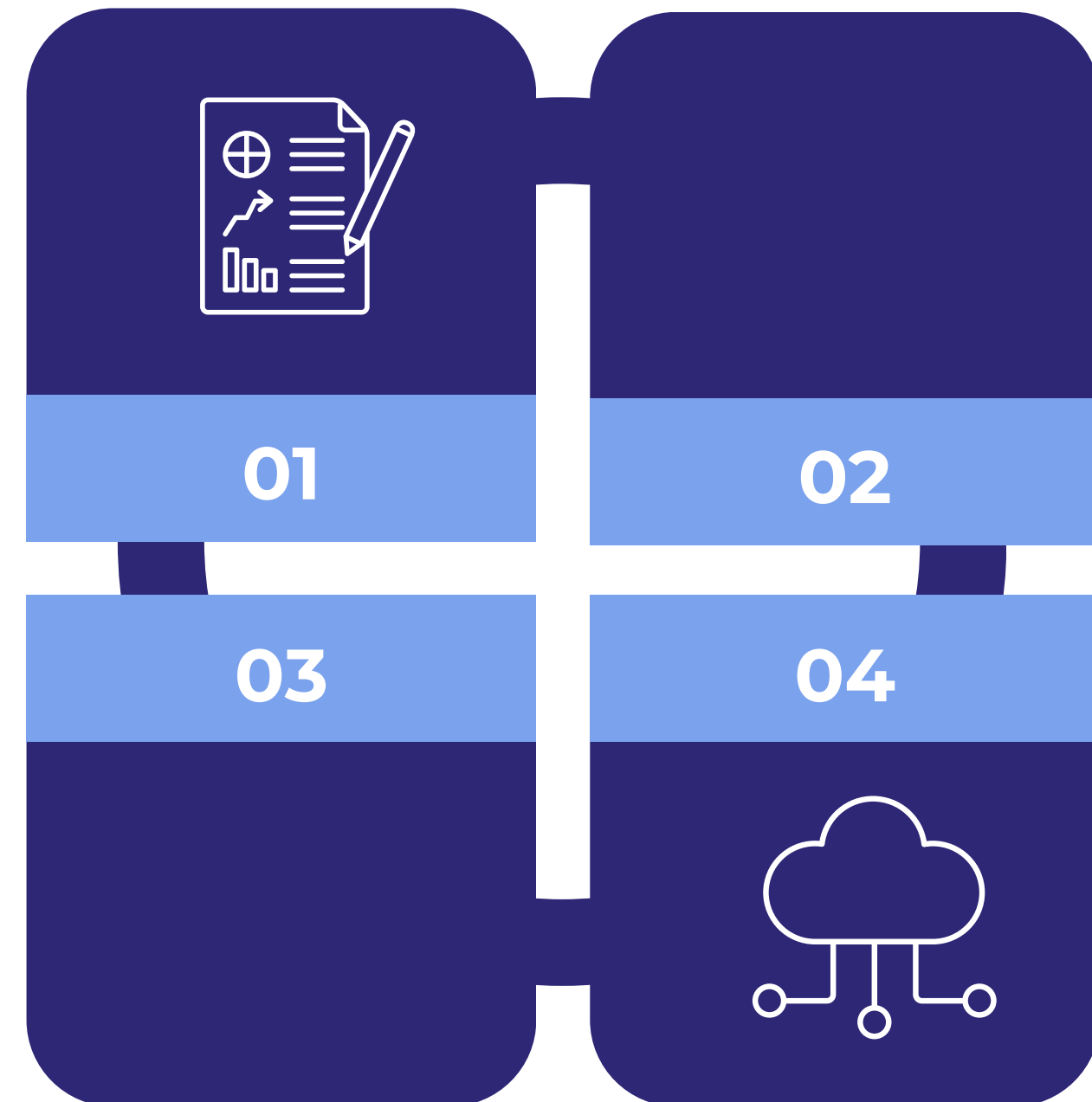
We comply with the standards

Context

Our focus for regulatory compliance will be on the **sensors**, and not the autonomous car as a whole.

Physical Car

We will be assuming that the underlying car (Chevrolet EUV Bolt) already complies with all the regulations.



MAIN REGULATORY BODIES



FEDERAL MOTOR VEHICLE SAFETY STANDARDS (FMVSS)

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset]



FMVSS 111: Regulation for **Rear Visibility Requirements**, including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

FMVSS 127: Regulation for **Automatic Emergency Braking** and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

FMVSS 208: Regulation for **Occupant Crash Protection** design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.



DMV REGULATIONS

NOTE: Only for the US [SAE AutoDrive R2Y5 Ruleset]

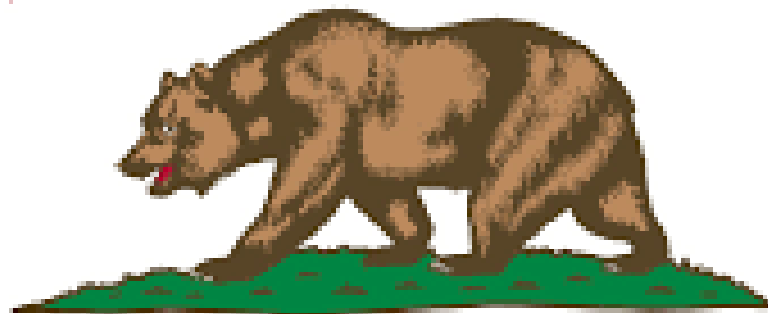


1. Provide an easily accessible method for the driver to engage or disengage the autonomous driving system.
2. Include an in-vehicle indicator that informs the driver when the autonomous driving system is active.
3. Have a system that alerts the driver if the autonomous driving system fails and either allows manual takeover or safely brings the vehicle to a complete stop.
4. Comply with all applicable Federal Motor Vehicle Safety Standards.
5. Include a separate mechanism to record and store sensor data from at least 30 seconds before a collision.



STATE OF CALIFORNIA

Clearing all DMV requirements is enough for California!

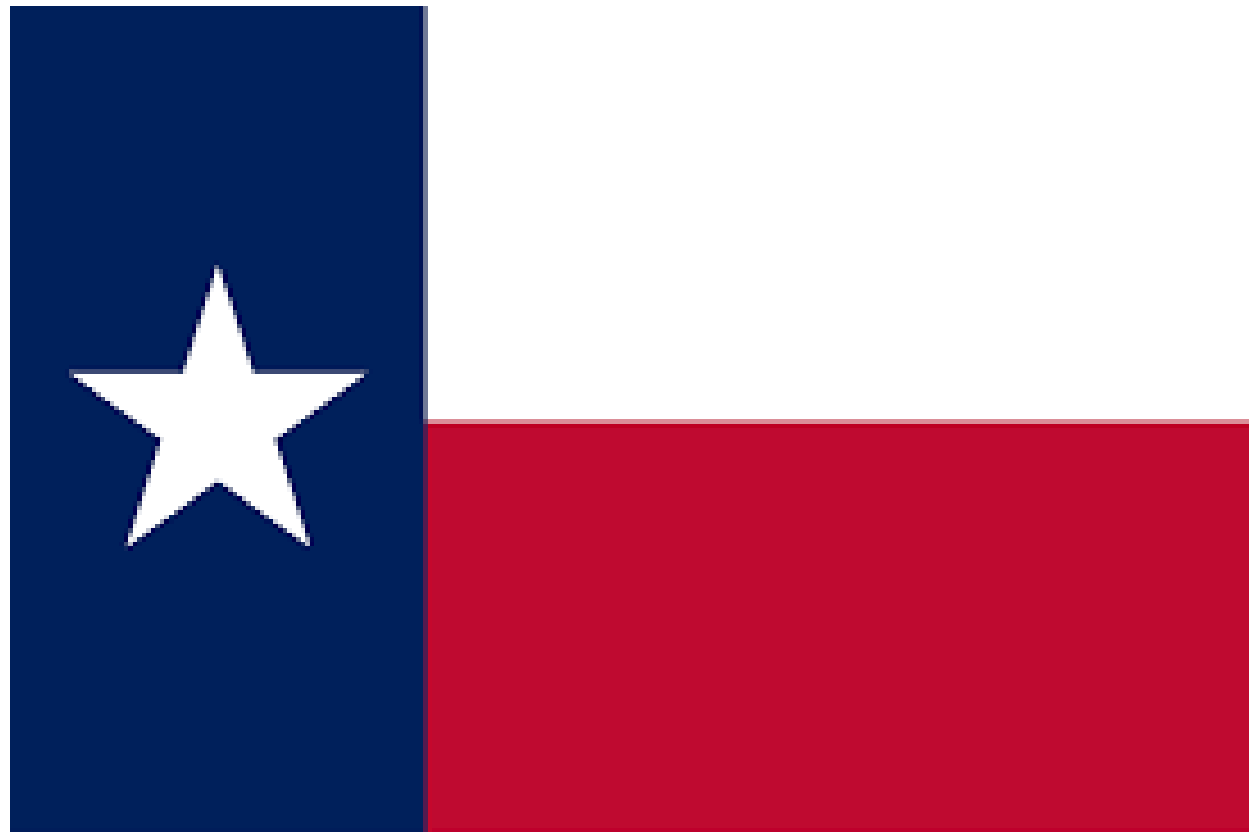


CALIFORNIA REPUBLIC



STATE OF TEXAS

Alongside clearing the DMV requirements...



1. Need to create a first responders safety plan (more on our self certification soon!)
2. The DMV must have enhanced authority to oversee autonomous vehicle operations
3. Need to report safety data and incidents [20]



STATE OF ARIZONA

Alongside clearing the DMV requirements...



1. Must submit a plan to ADOT and the Arizona Department of Public Safety
 - explains how law enforcement should interact with your autonomous vehicles during emergencies, traffic stops, and accidents.
2. A self certification acknowledging:
 - Compliance with all federal safety standards.
 - Compliance with all Arizona traffic and motor vehicle safety laws. [23, 24, 25]
 - can achieve a “minimal risk condition” (i.e., safely stopping) in case of system failure.



STATE OF MICHIGAN

Alongside clearing the DMV requirements...



1. Must obtain special manufacturer plates issued by the Michigan Secretary of State.
2. Must report proof of insurance, business presence in Michigan, and details about the autonomous system to the Secretary of State. [21, 22]



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
- It contains an indicator inside of the vehicle that notifies the driver when the autonomous driving system is engaged;
- It has a means to alert the driver when the autonomous driving system has failed and either allows the driver to take manual control of the vehicle, or is capable of bringing itself to a complete stop;
- It meets all applicable Federal Motor Vehicle Safety Standards; and
- It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.

Our R2Y4 car (Artemis) already has the feature!



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
- **It contains an indicator inside of the vehicle that notifies the driver when the autonomous driving system is engaged;**
- It has a means to alert the driver when the autonomous driving system has failed and either allows the driver to take manual control of the vehicle, or is capable of bringing itself to a complete stop;
- It meets all applicable Federal Motor Vehicle Safety Standards; and
- It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.

Artemis has lights inside and outside of the car to indicate what mode it is in.



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
- It contains an indicator inside of the vehicle that notifies the driver when the autonomous driving system is engaged;
- **It has a means to alert the driver when the autonomous driving system has failed and either allows the driver to take manual control of the vehicle, or is capable of bringing itself to a complete stop;**
- It meets all applicable Federal Motor Vehicle Safety Standards; and
- It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.



Our additional features of our MVP accommodates for this!



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
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- **It meets all applicable Federal Motor Vehicle Safety Standards; and**
- It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.



The Chevrolet EuV Bolt complies with these standards!



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
- It contains an indicator inside of the vehicle that notifies the driver when the autonomous driving system is engaged;
- It has a means to alert the driver when the autonomous driving system has failed and either allows the driver to take manual control of the vehicle, or is capable of bringing itself to a complete stop;
- It meets all applicable Federal Motor Vehicle Safety Standards; and
- **It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.**



Our additional features of our MVP accommodates for this!



DMV REGULATIONS

- It is equipped with an easily accessible way for the driver to engage or disengage the autonomous driving system;
- It contains an indicator inside of the vehicle that notifies the driver when the autonomous driving system is engaged;
- It has a means to alert the driver when the autonomous driving system has failed and either allows the driver to take manual control of the vehicle, or is capable of bringing itself to a complete stop;
- **It meets all applicable Federal Motor Vehicle Safety Standards; and**
- It has a separate mechanism to capture and save the vehicle's sensor data at least 30 seconds before a collision involving the vehicle occurs.

The Chevrolet EuV Bolt complies with these standards!



STATE OF ARIZONA

The state of Arizona also does not require a DMV permit, but require other specific compliance requirements:

- Must submit a plan to ADOT and the Arizona Department of Public Safety that explains how law enforcement should interact with your autonomous vehicles during emergencies, traffic stops, and accidents.
- Must provide a self certification acknowledging:
 - The vehicle complies with all federal safety standards and any NHTSA exemptions are disclosed.
 - The vehicle can achieve a “minimal risk condition” (i.e., safely stopping) in case of system failure.
 - The vehicle will fully comply with all Arizona traffic and motor vehicle safety laws.

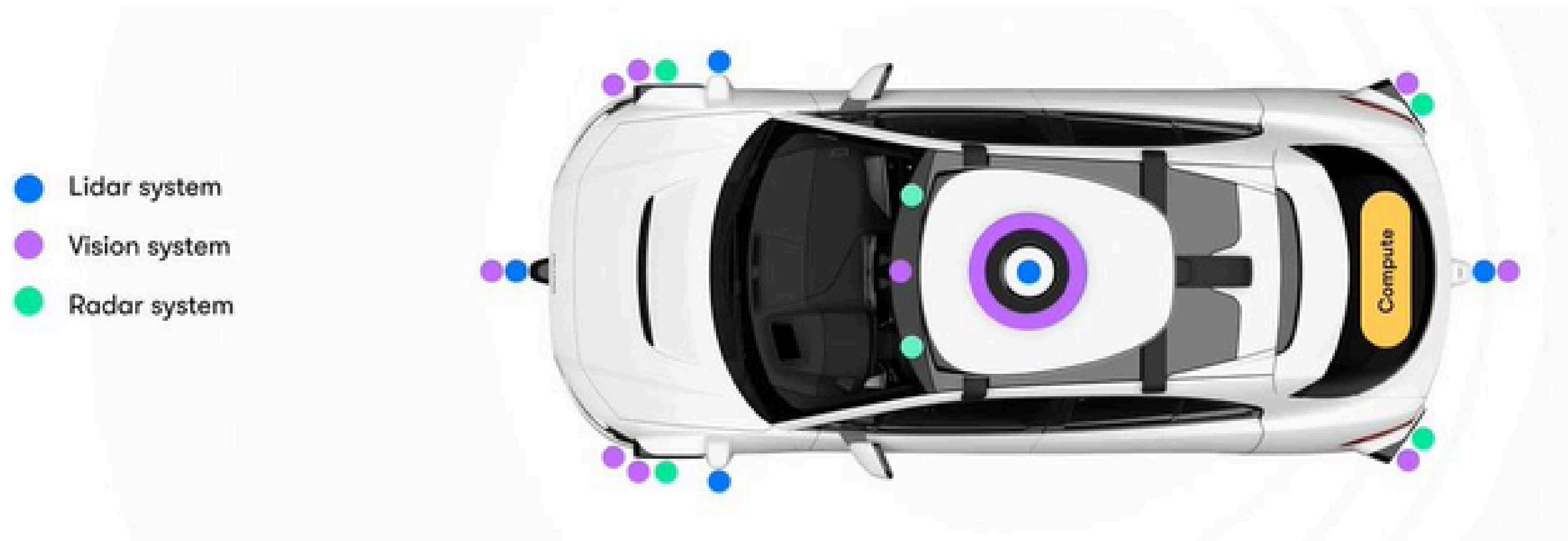


MVP



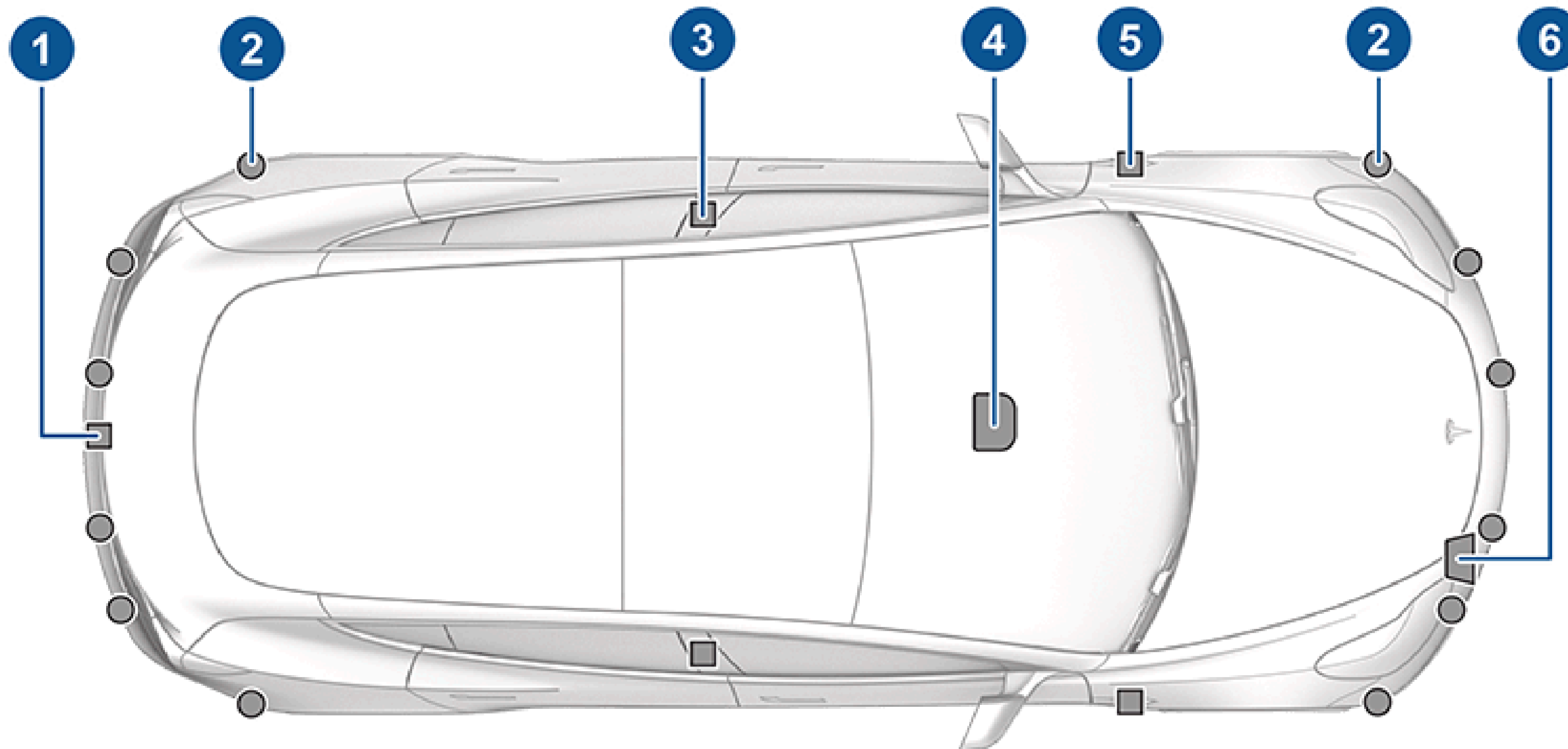
INSPIRATION

Waymo's sensor suite:

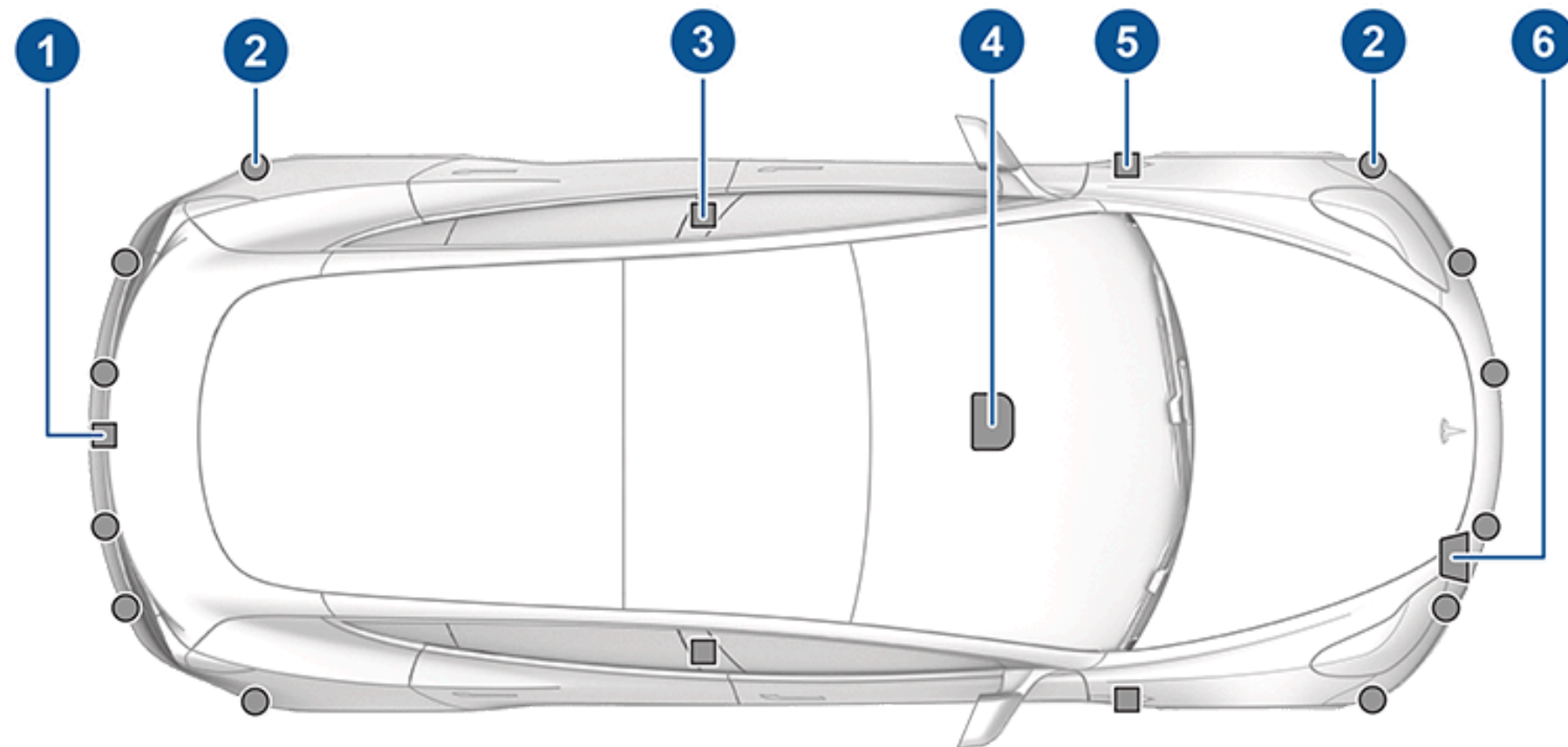


INSPIRATION

Tesla's sensor suite:
#6 is radar, #2 is ultrasonic



Tesla's sensor suite



6 is radar, and 2 is ultrasonic.

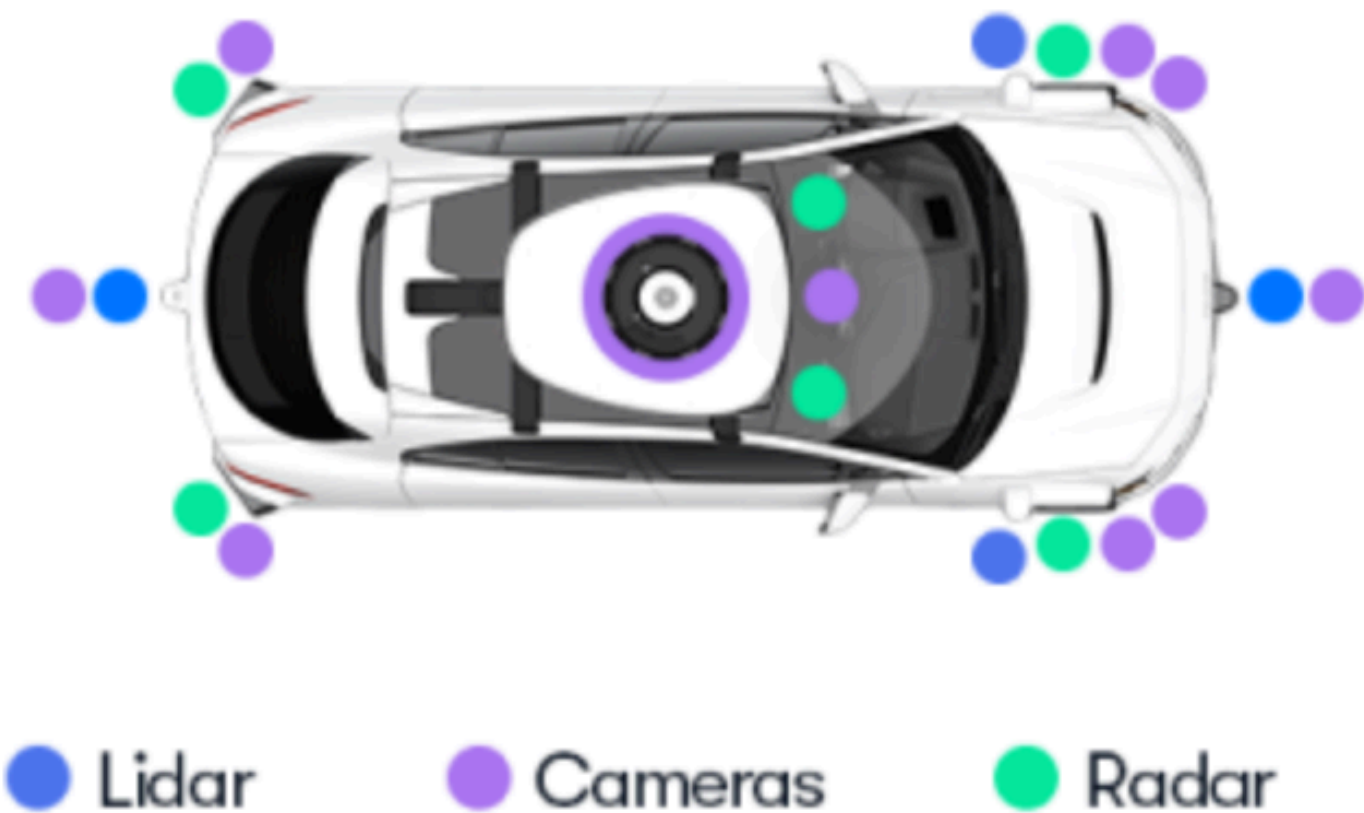
Tesla's Fully Self-Driving
functionality relies
completely on cameras.

NOTE: Tesla's FSD is only L2-
L3 [9]



Waymo's Sensor Suite [10, 11, 12]

Jaguar I-PACE



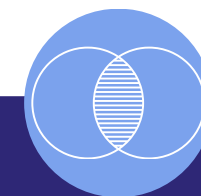
Sensor Modality	Quantity in 6th Gen	Power Consumption (Watts)
Cameras	13	2–5W per unit
LiDAR (Top 360°)	1	12–20W
LiDAR (Perimeter)	3–4	8–12W per unit
Radar	6	3–8W per unit
Ultrasonic	8–12	0.05–0.15W per unit



DESIGN PRINCIPLES AND CONSTRAINTS



Camera heavy for cost and power efficiency.



Having minimal sensor overlap, but not decreasing functionality.



Leveraging LiDAR and radar to target L4-L5 autonomy.



Compliant with all the regulations in our chosen locations.



LiDAR sensors on the market

<u>Sensor</u>		<u>Metrics</u>			
Brand	Model	Range (m)	H x V RES (degrees)	H x V FOV (degrees)	Price
Luminar	Model J+	250	0.05 x 0.05	120 x 26	650
Koito/Cepton	Cepton Plus	250	0.05 x 0.05	120 x 25	600
Aeva	Atlas 8-Beam	220	0.044 x 0.025	120 x 25	575
Luminar	Iris I	250	0.1 x 0.05	120 x 26	800
Innoviz	InnovizTwo-Osprey	210	0.05 x 0.05	120 x 28.8	700
Hesai	AT128	210	0.1 x 0.2	120 x 25.4	~400
Hesai	ATX	200	0.08° x 0.1°	120 x 20	~200
Luminar	Iris+	280	0.05 x 0.05	120 x 26	~1300
Ouster	OS1	200	0.18 x 0.7	360 x 42.4	~2000

The sensors on the market

CAMERA

Sensor	Metrics		
Brand	H x V FOV (degrees)	Price	Resolution (MP)
Valeo	195 x 155	21	1
Valeo	196 x 155	35	2
Continental	197 x 155	21	1
Continental	198 x 155	35	2

RADAR

Sensor		Metrics		
Brand	Model	Range (m)	H x V FOV (degrees)	Price
Hella	SRR5G3	90	75 x 10	35
Veoneer	LRR120	193	45 x 9	50
Continental	ARS4G0	250	60 x 9	150
Continental	SRR520	100	75 x 7.5	75
Continental	ARS513	220	45 x 9	80



The sensors on the market

Cameras are one of the **cheapest** sensors available

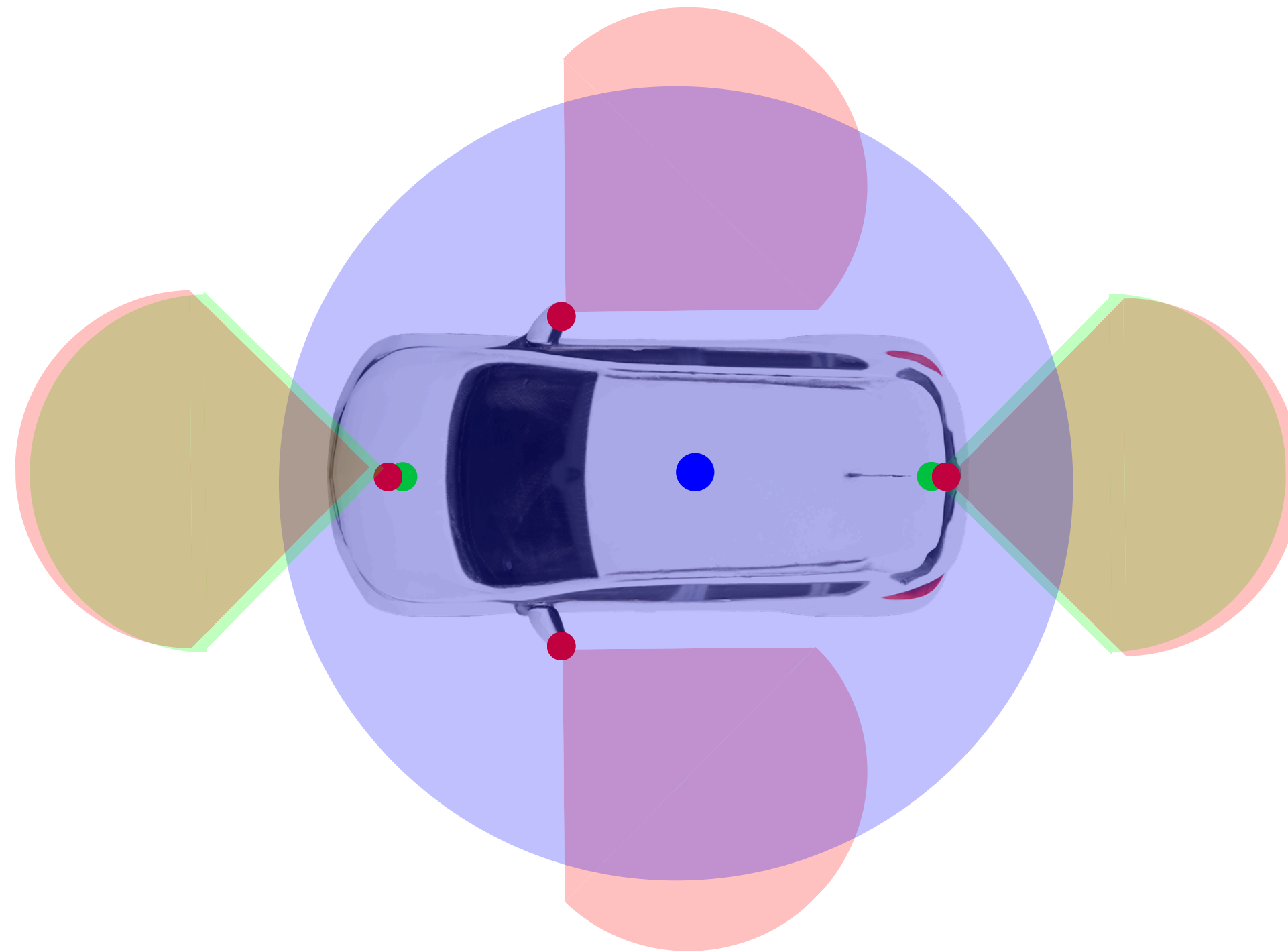


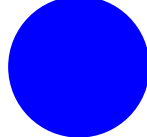
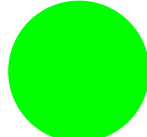
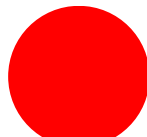
Sensor	Metrics		
Brand	H x V FOV (degrees)	Price	Resolution (MP)
Valeo	195 x 155	21	1
Valeo	196 x 155	35	2
Continental	197 x 155	21	1
Continental	198 x 155	35	2

Our sensor suite
will be camera
heavy



Our MVP

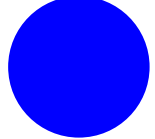
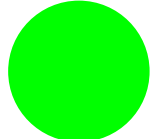
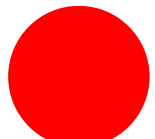


-  Lidar
-  Radar
-  Camera



Our MVP

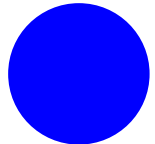
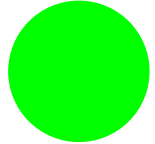
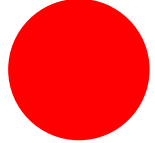


-  Lidar
-  Radar
-  Camera



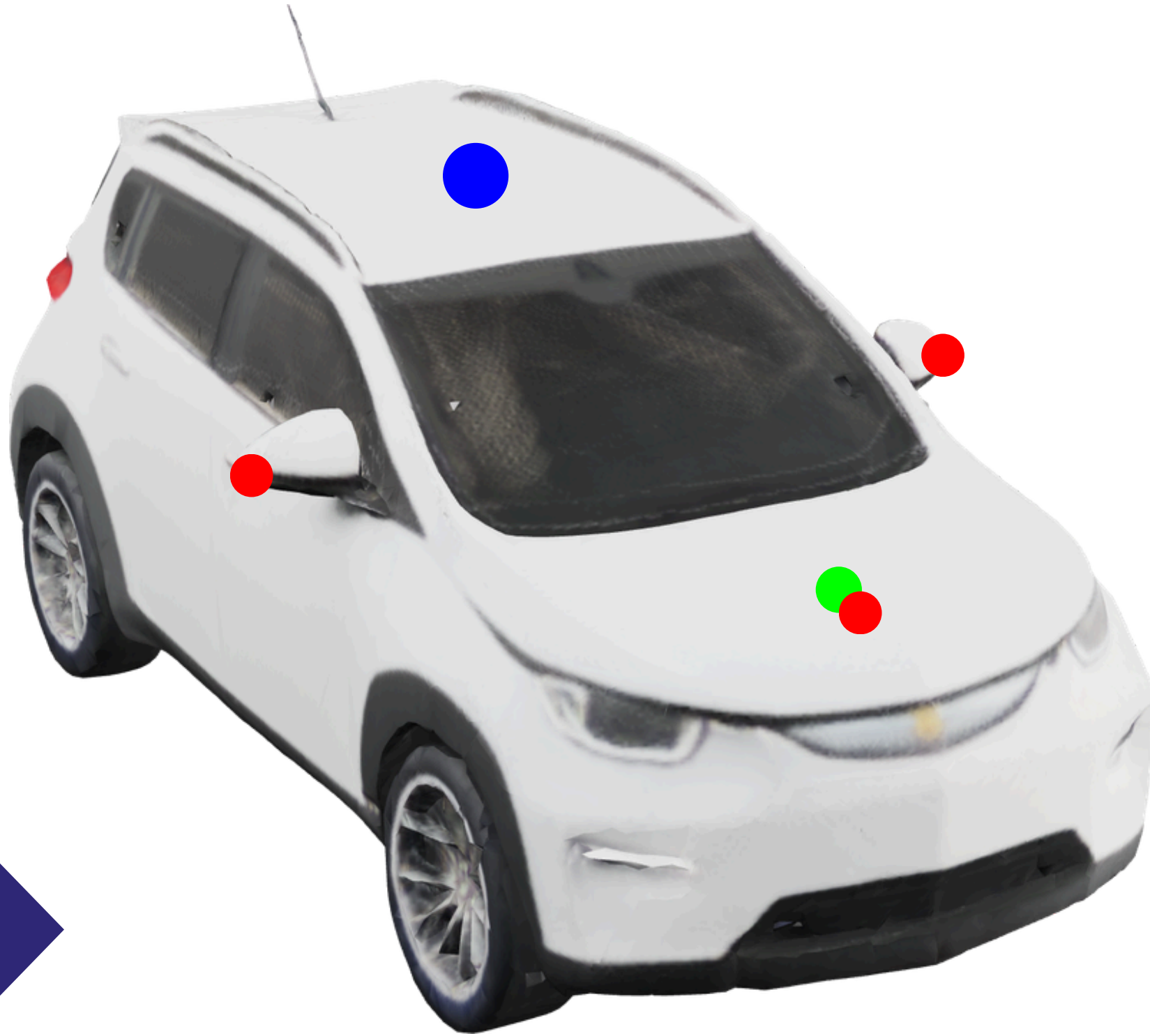
Our MVP



-  Lidar
-  Radar
-  Camera



Our MVP



Extra Additional Features:

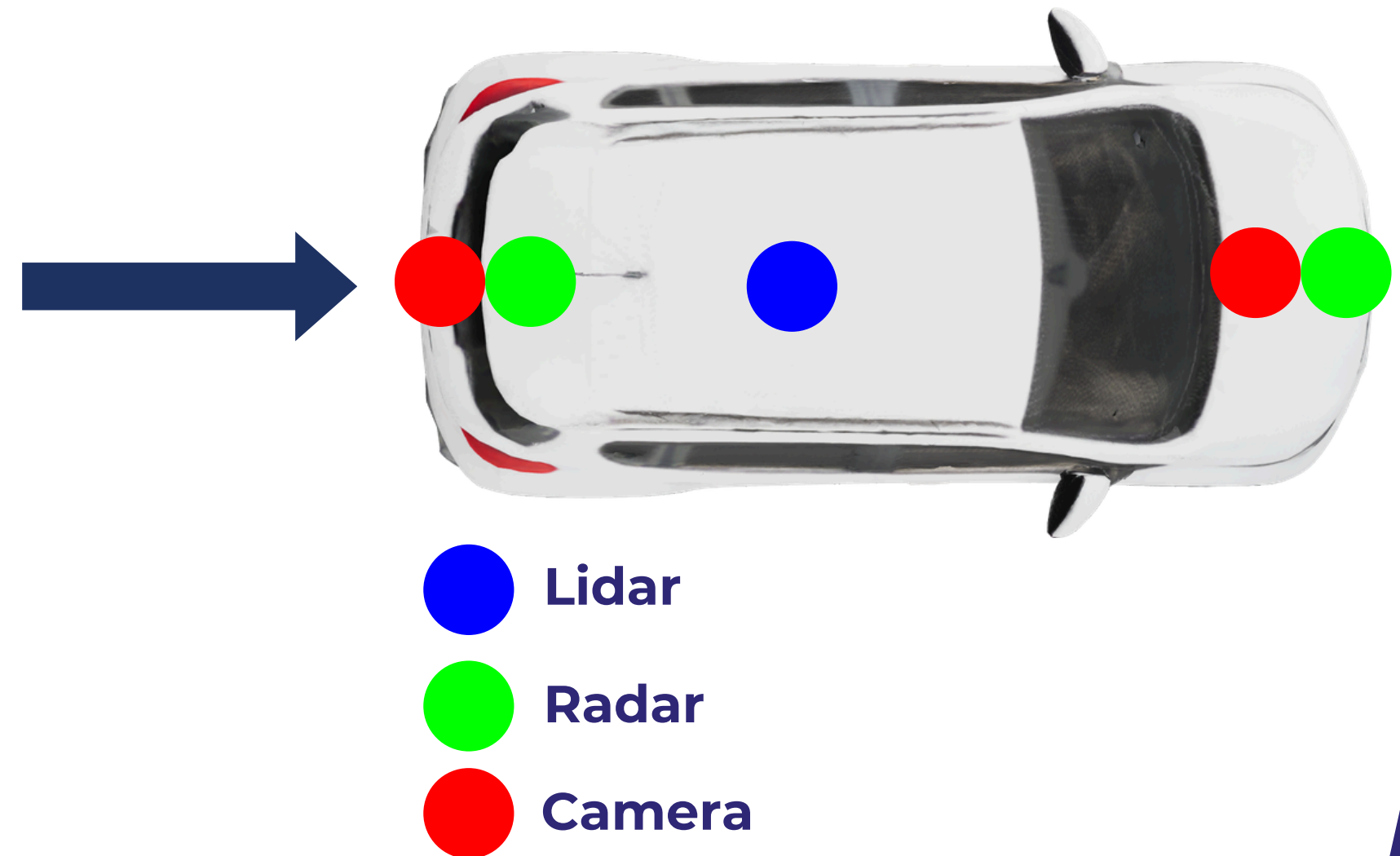
1. All the sensor data 1 minute before a crash is saved.
2. There will be an internal alerting system to indicate to the driver to take control again.



FMVSS 111: REAR VISIBILITY

FMVSS 111: Regulation for Rear Visibility Requirements, including equipment of rearview camera that provides a clear field of view behind the vehicle when in reverse.

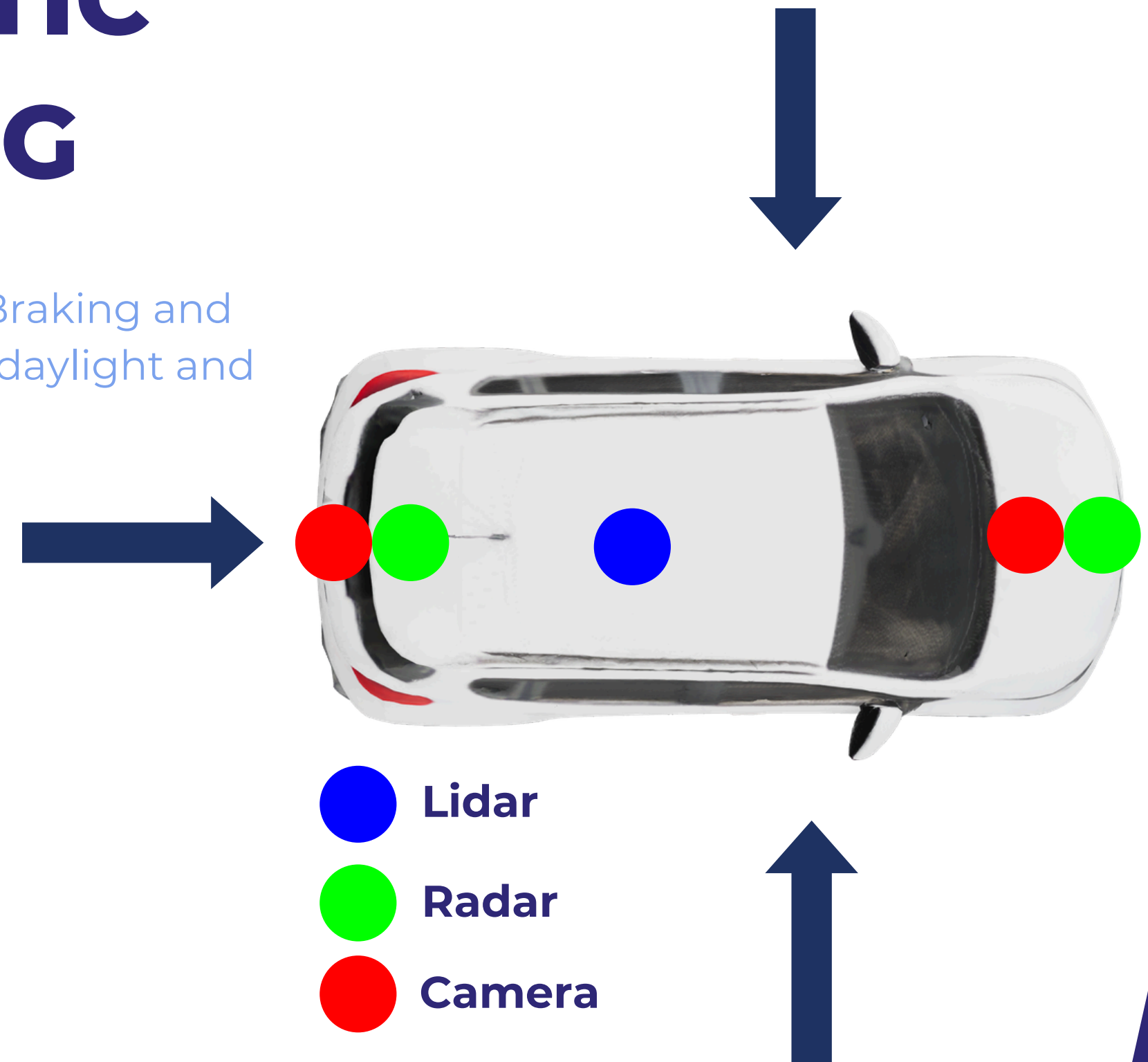
Our sensor suite has 3 modalities of sensors for rear visibility which is more than enough to comply with this regulation, as the minimum requirement is just a clear field of view.



FMVSS 127: AUTOMATIC EMERGENCY BRAKING

FMVSS 127: Regulation for Automatic Emergency Braking and Pedestrian Automatic Emergency Braking in both daylight and darker conditions at night.

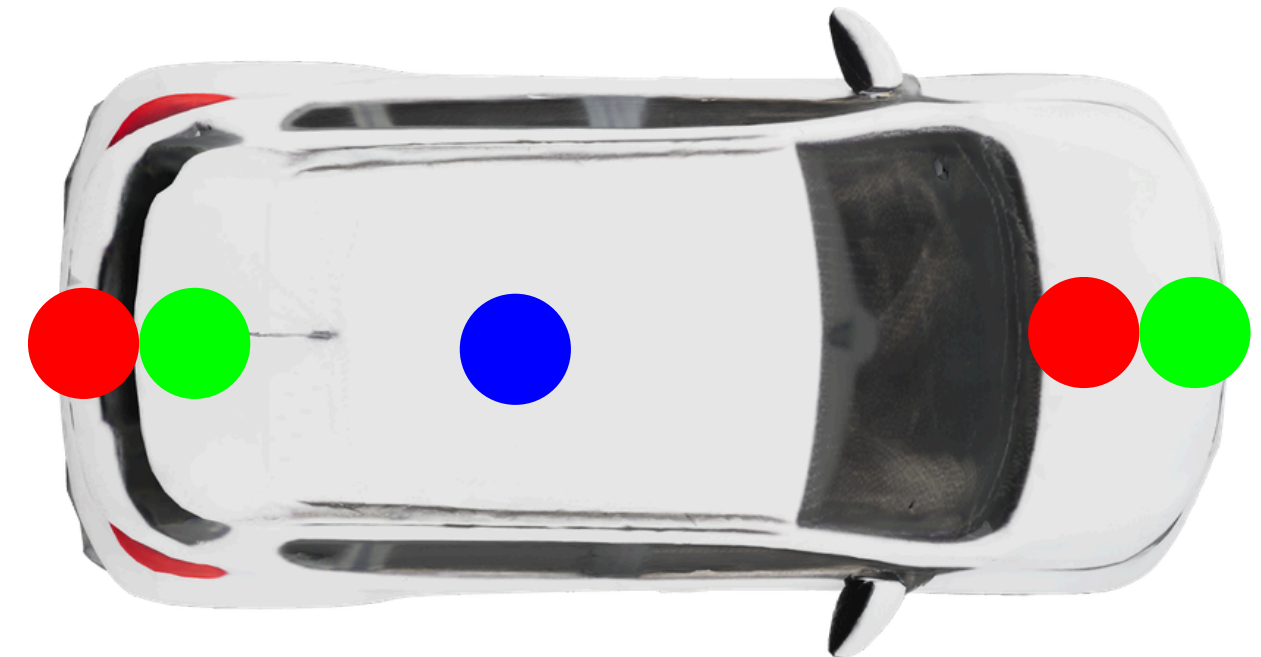
Radar sensors are best for detection during darker/misty/low visibility scenarios, and we have 75 degrees facing the front and back.



FMVSS 208: OCCUPANT CRASH PROTECTION

FMVSS 208: Regulation for Occupant Crash Protection design and performance of safety systems in motor vehicles. Includes seat belts, airbags, crash testing, and child restraint systems.

Our sensor suite has nothing to do with this regulation, as its regarding the functionality of the car itself. By our assumption, we will comply with this regulation.



KEY SPECS



Total Energy
Consumption: **72W**



Old Mileage:
259 miles



Total Cost of Sensors:
\$2154

New Mileage:
231 Miles



SENSOR DESIGN CHOICES

Cost and Power Consumption

LiDAR: ~ \$2,000*, ~17 W

2x Radar: ~ \$35, 22.5 W**

4x Camera: ~ \$21, 2.5 W**

* Lidar price estimated from a listed device on sale

** Taken from R2Y4 Triple Zero



Our Sensors

Valeo cameras 4x

Brand	H x V FOV (degrees)	Price	Resolution (MP)	Power (W)
Valeo	195 x 155	21	1	<2.5

Hella radars 2x

Brand	Model	Range (m)	H x V FOV (degrees)	Price	Power (W)
Hella	SRR5G3	90	75 x 10	35	22.5

Ouster OS1 LiDAR

Brand	Model	Range (m)	H x V RES (degrees)	H x V FOV (degrees)	Price	Power (W)
Ouster	OS1	200	0.18 x 0.7	360 x 42.4	~2000	14-20



New Mileage [13, 14]

Total upfront cost of sensor suite: **\$2154**

The mileage of the Chevrolet Bolt EV: **259 miles**

Total energy consumption (60mph):

~72 W ->

0.0012 kWh/mile

The Chevrolet Bolt EV's battery capacity: **65 kWh**

Our new mileage: $65 / 0.2812 = 231.1522$ miles

The Chevrolet Bolt EV has an energy usage of:

0.28 kWh/mile

~231 miles

The Chevrolet Bolt with the MVP:

$0.28 \text{ kWh/mile} + 0.0012 \text{ kWh/mile}$

= 0.2812 kWh/mile



Our MVP

50

Comparing our MVP



Self-Certification Plan & ODD Sensor Compliance



ODD HIGH LEVEL OVERVIEW



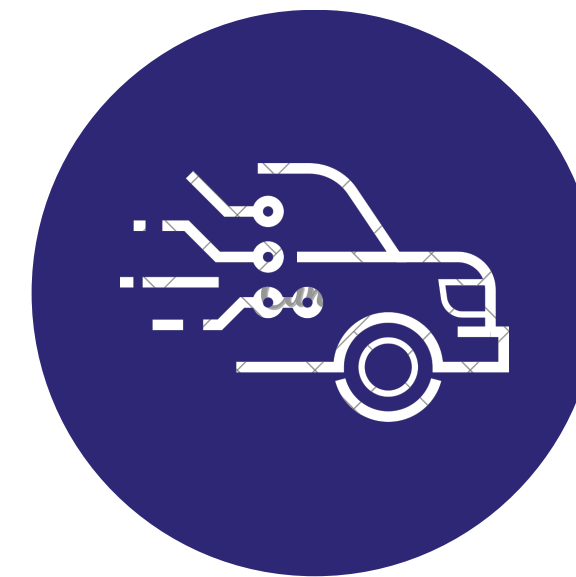
Environmental

Weather
Geolocation
Object Avoidance



Public roads & Traffic information

Parking
Road Types
Congestion
Road Grade



Autonomy

Path Planning & Execution
Lateral Controls
Longitudinal Controls



Our MVP

ENVIRONMENTAL

ODD and Self Certification

Weather

Conduct controlled track tests (10–30 trials/condition),

Geolocation

Conduct tests across different map types (interstate/arterial/urban)

Object Avoidance

Place standardized obstacles and run trials at multiple speeds (20–65 km/h) and lighting conditions



Our MVP

p

PUBLIC ROADS, TRAFFIC INFORMATION

ODD and Self Certification

Road Types

Run simulation, then a multi-turn track test for straight, curve, and roundabout segments.

Congestion

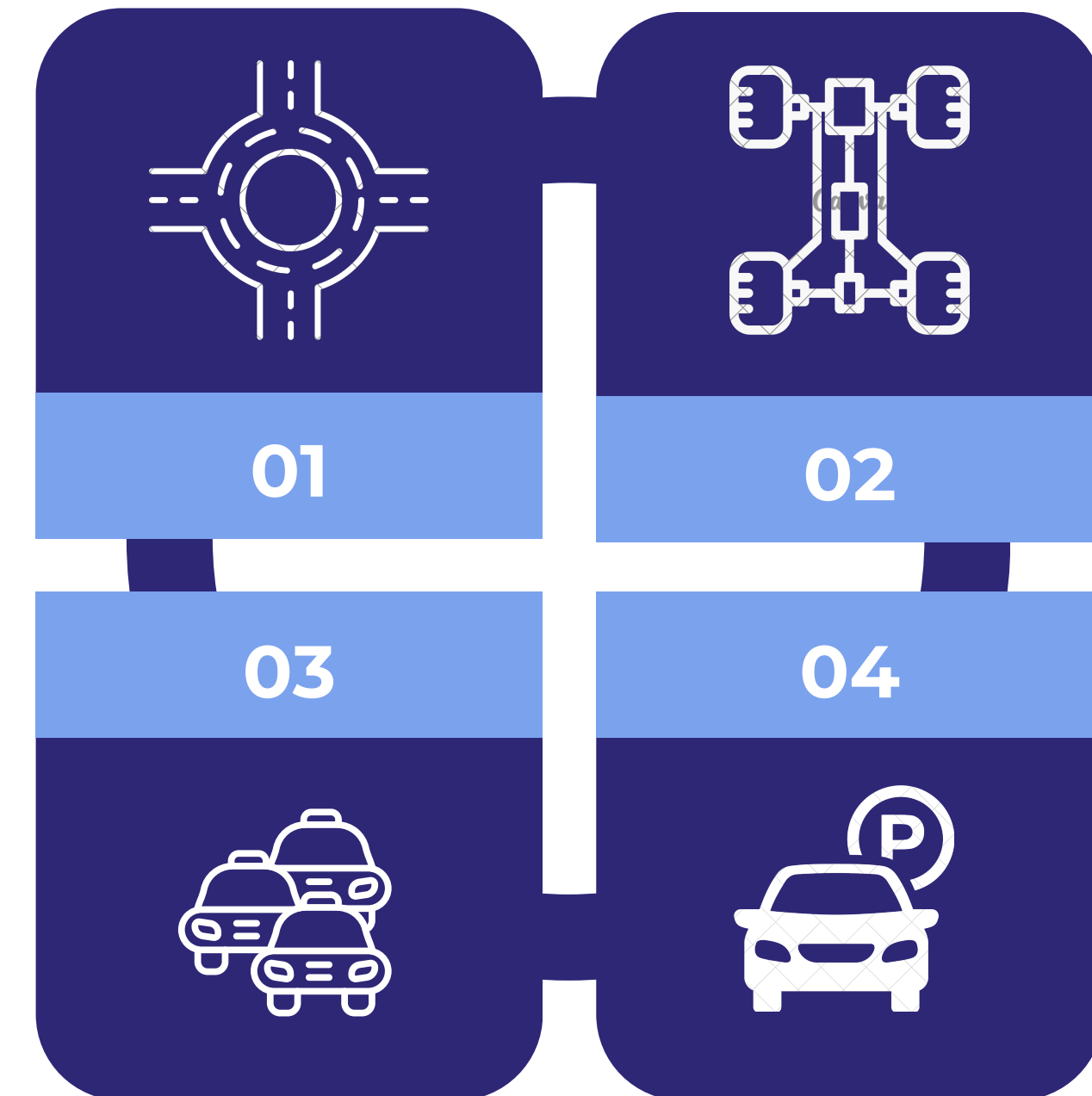
Run simulation, then create a test suite to simulating light to heavy congestion with principal other vehicles

Road Grade

Set up chassis dyanometer device to simulate ascents/descents at fixed speeds.

Parking

Create standard stalls for all parking categories, perform simulation tests then on track for forward & reverse parking



Our MVP

p

AUTONOMY

ODD and Self Certification

Longitudinal Controls

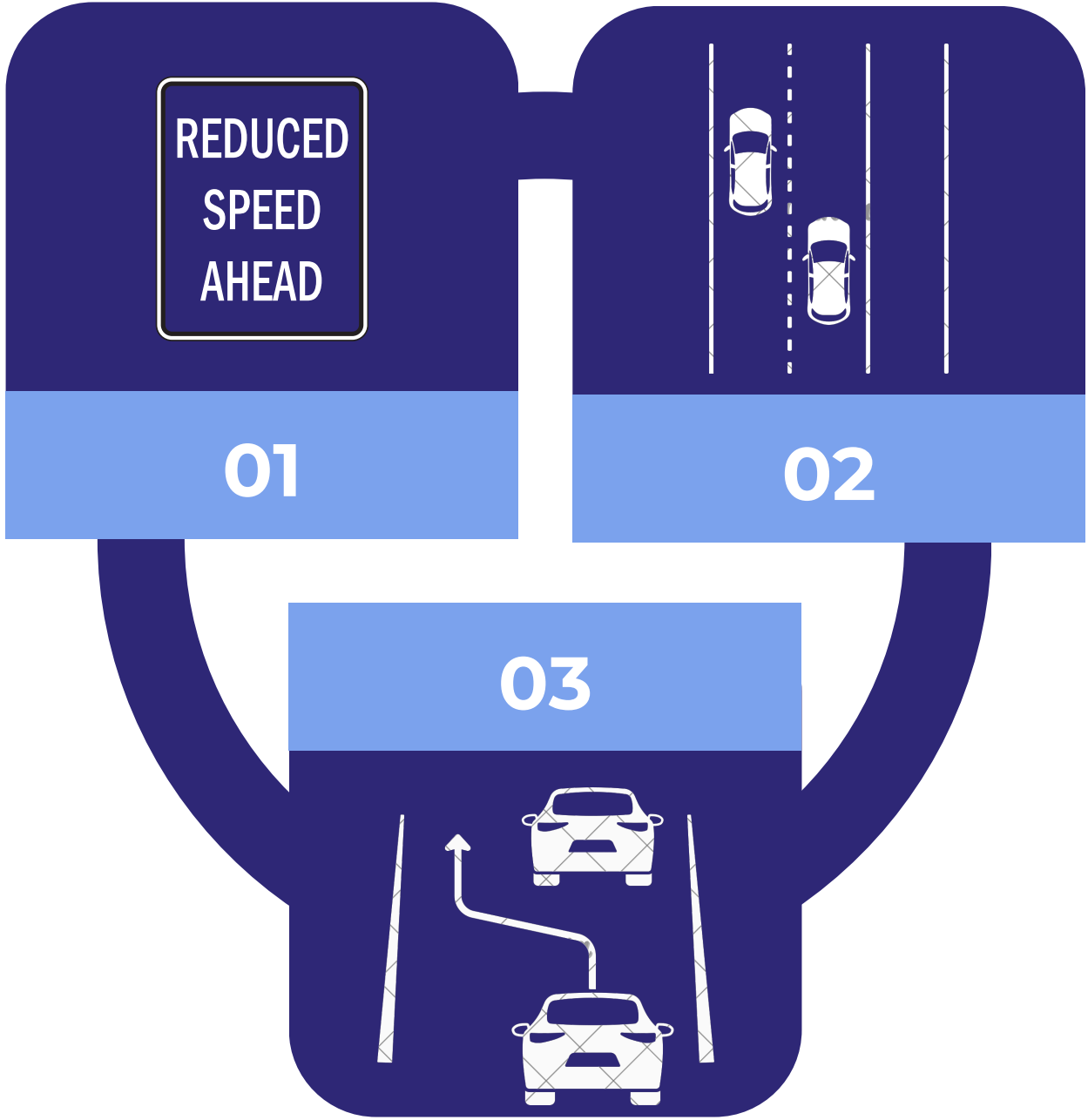
Verify in simulation that the vehicle can safely follow, detect signs, and adjust speed before track testing.

Lateral Controls

Verify in simulation vehicle's lane centering, lane changes, and turn capabilities in simulation before track testing

Path Planning & Execution

Run simulation, then a multi-turn track test with lane changes, merges, detour, random signals, and emergency-vehicle events



Our MVP

p

How did we connect ODD-Level Tests to System Certification?



SELF-CERTIFICATION PLAN

We validate system compliance with:

FMVSS 111

FMVSS 127

FMVSS 208

We validate operational compliance with the following ODDs:

Environmental

Weather
Geolocation
Object Avoidance

Public roads & Traffic information

Parking
Road Types
Congestion
Road Grade

Autonomy

Path Planning & Execution
Lateral Controls
Longitudinal Controls

This certification is supported by:

- Verified ODD-level test evidence
- Sensor coverage and performance validation
- Simulation, track, and real-world testing



CHALLENGES & KEY LEARNINGS

Presentations are tools that can be used as lectures, speeches, reports, and more. It is mostly presented before an audience.



01.Advantage

It is mostly presented before an audience.



02.Advantage

It is mostly presented before an audience.



03.Advantage

It is mostly presented before an audience.



THE TEAM

THANK YOU

Jamie Chastain

Manager



Isabel Mercado

Manager



Shawn Garcia

Manager



Yanis Petros

Manager



Our MVP

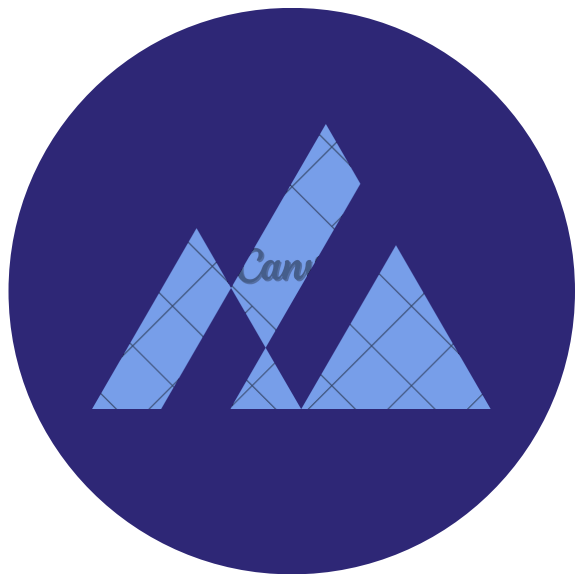
TITLE

Test Execution → Data Analysis
→ Compliance Documentation



Competitor 01

It is mostly presented
before an audience.



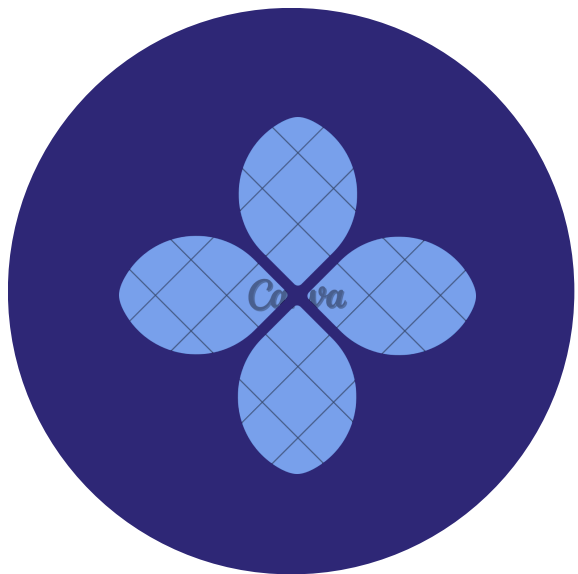
Competitor 02

It is mostly presented
before an audience.



Competitor 03

It is mostly presented
before an audience.



Competitor 04

It is mostly presented
before an audience.



Our MVP

MARKET SIZE

Presentations are tools that can be used as lectures, speeches, reports, and more. It is mostly presented before an audience.



\$500 M

Total Available Market

It is mostly presented
before an audience.



\$100 M

Serviceable Available Market

It is mostly presented
before an audience.



\$50 M

Serviceable Obtainable Market

It is mostly presented
before an audience.



Our MVP